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Manual for XenoBase™:

XenoBase™ Functionality Summary:

1. Pharmacology: XenoBase™ allows you to examine the pharmacology information of cell lines under different models and drugs.
2. Gene Expression: XenoBase™ allows you to compare the expression level of a single gene (or simultaneously compare upto eight genes) on a selection of cell lines.
3. Gene Copy Number: XenoBase™ allows you to examine the gene copy number of a single gene over a selected number of cell lines; you can also examine the gene copy number landscape over the whole genome (for upto 100 cell lines at a time), or over the corresponding chromosome of a selected gene.
4. Gene Mutation: XenoBase™ allows you to examine the mutation status of selected genes.

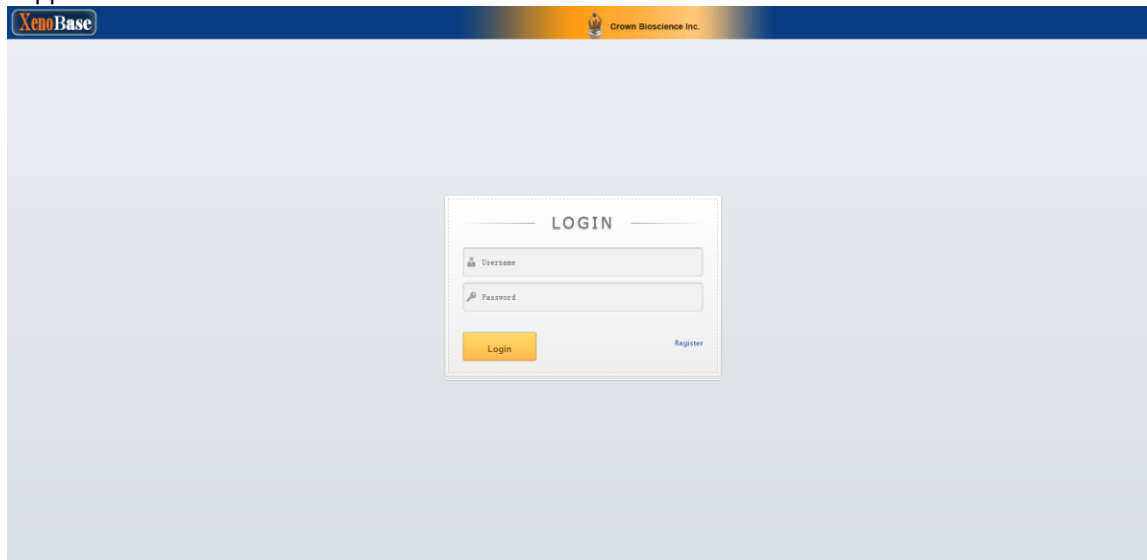
This manual is written for the first time users of XenoBase™, and tells users how to perform the queries step by step.

The content of this manual is organized by type of queries. One query type corresponds to one subtitle.

To ensure maximal functionality is displayed, please use Internet Explorer and update to the latest version of Adobe Reader when logging into XenoBase™.

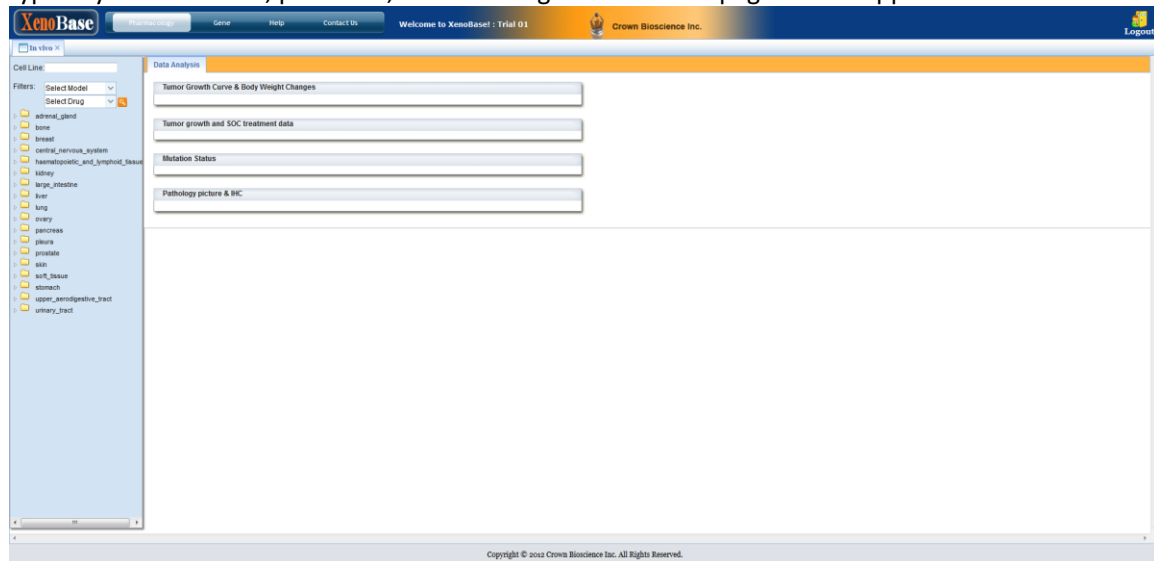
1. Log in

Type "http://58.210.198.254:8080/" into the address bar of the IE browser, the login page should appear as below:



The screenshot shows the XenoBase login interface. At the top, there is a dark blue header bar containing the 'XenoBase' logo on the left and 'Crown Bioscience Inc.' on the right. Below the header, the main background is a light blue gradient. Centered on this background is a white rectangular login box. Inside the box, the word 'LOGIN' is centered at the top. Below it are two input fields: the first is labeled 'Username' with a small user icon to its left, and the second is labeled 'Password' with a small key icon to its left. At the bottom of the box, there is a yellow 'Login' button and a blue 'Register' link.

type in your username, password, and click "log in". the home page should appear as below:



2.Register

If you do not have an account, click the "register" in the log in page to create one:

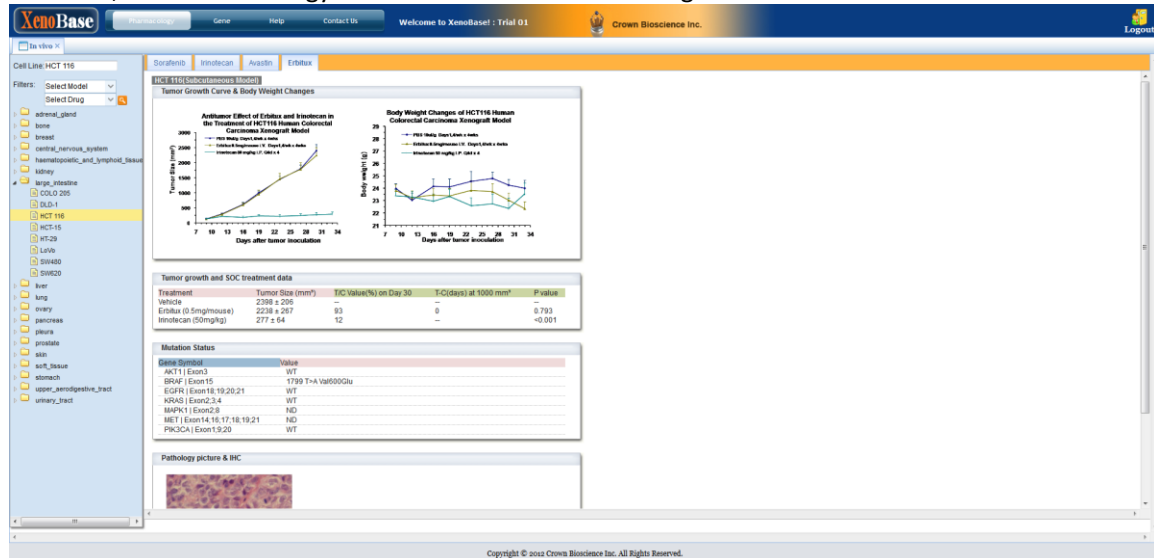


User register	
User Name:	
User Password:	
Email:	
Remark:	
register reset	

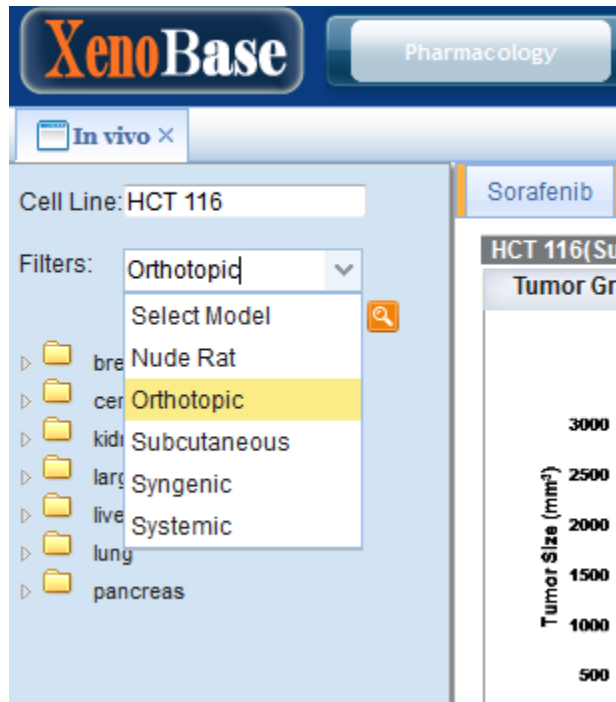
Type in your account name, password, and email address, then click "register" to submit your information (our administrator will review your application before activating your account). A dialog box will pop up to inform you that the registration is successfully submitted:

3. Pharmacology-In vivo

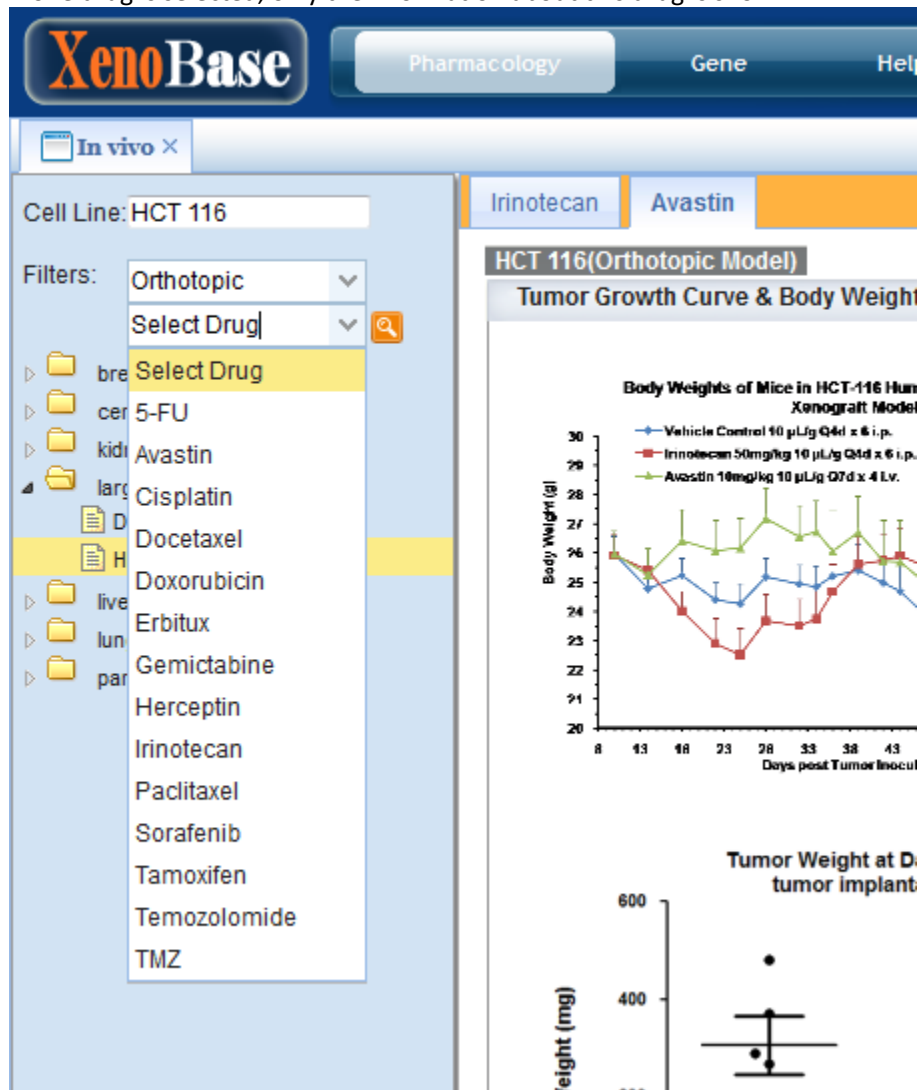
select one cell line after the prompt of "cell line" and set Filters (select model and drug) below, then click , the Pharmacology information will show at the right side.



If filters were left unselected, then information about this cell line on all the drugs and model will be shown.

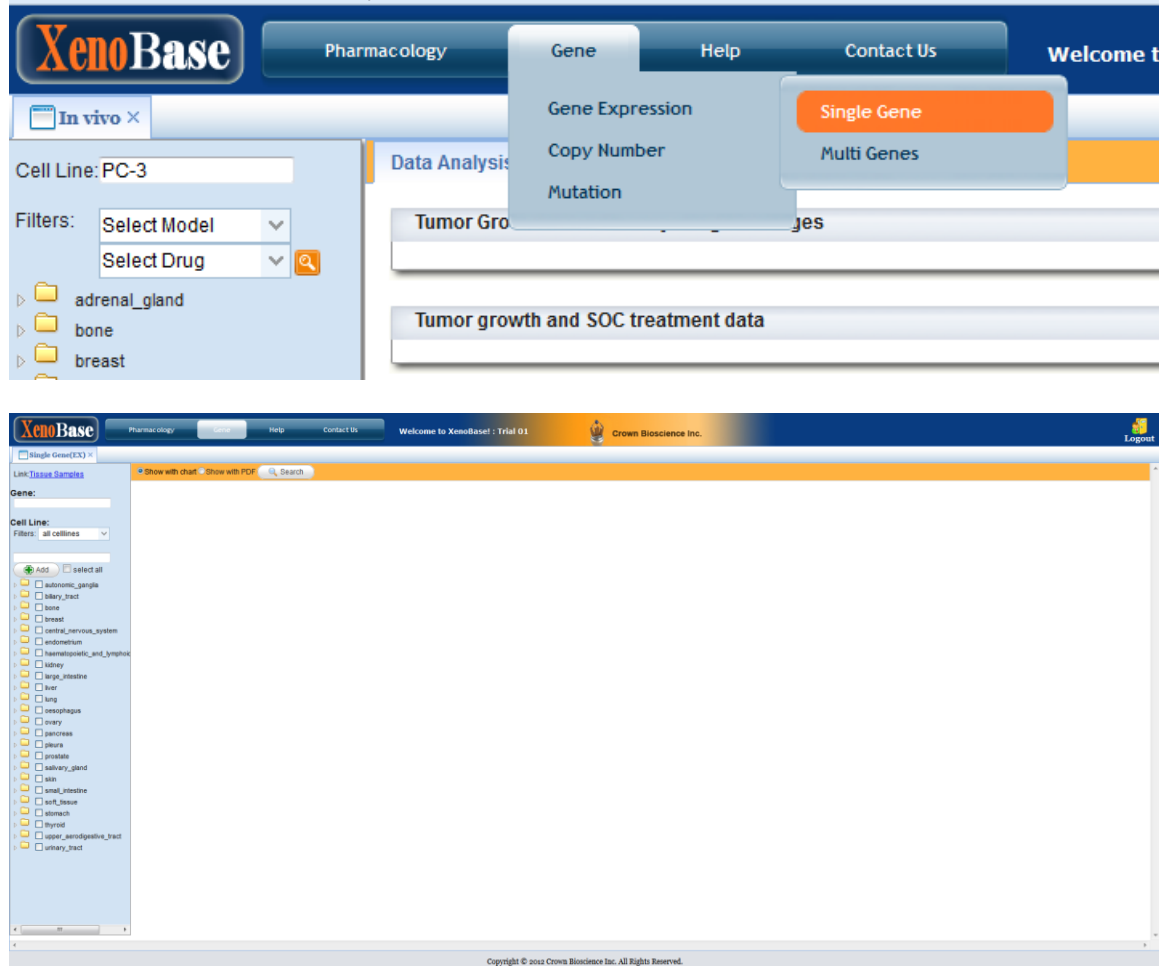


If one drug is selected, only the information about this drug is shown.



4. Gene-Gene Expression-Single Gene

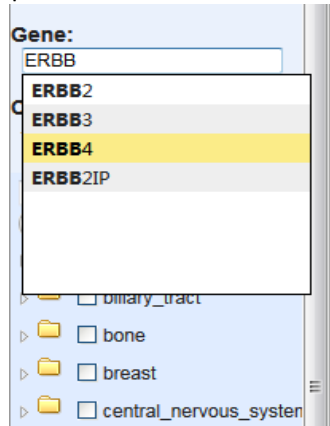
Click the "single gene" item from the navigation bar:



You may click "Tissue Samples" to jump to an outside website to look for gene expressions in primary tumors.

4.1 selecting gene and cell lines:

Type in gene names under the prompt of "Gene:"

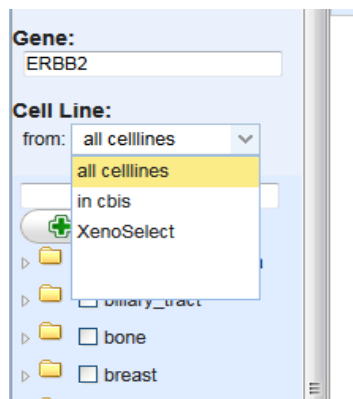


Select the cell line category under the prompt of "Cell Line:"

all celllines: includes all the cell lines in the CCLE database.

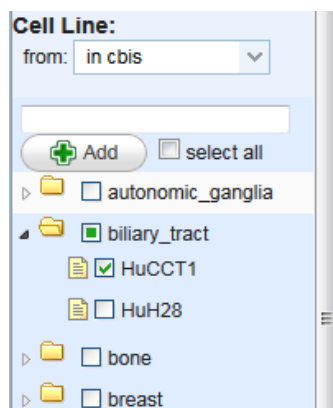
in cbis: includes the cell lines in the CCLE database and are maintained by CrownBio.

XenoSelect: includes the cell lines in the CCLE and are offered by CrownBio as validated xenograft models: ValidatedXeno®.

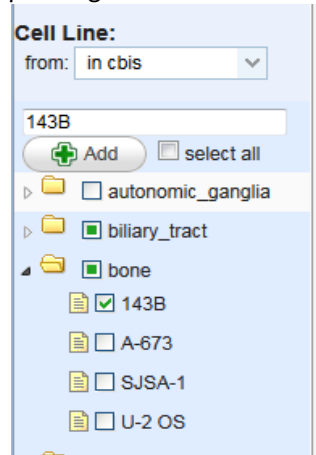


Then select cell lines through the inputbox or the checkbox:

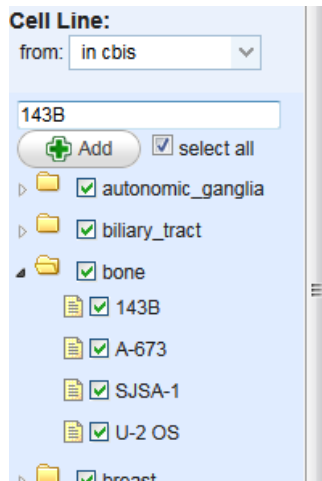
1. by checkbox:



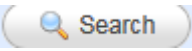
2. by entering the names of cell lines directly: for example, type in "143B" in the input box and click  , then the corresponding cell lines will be selected.



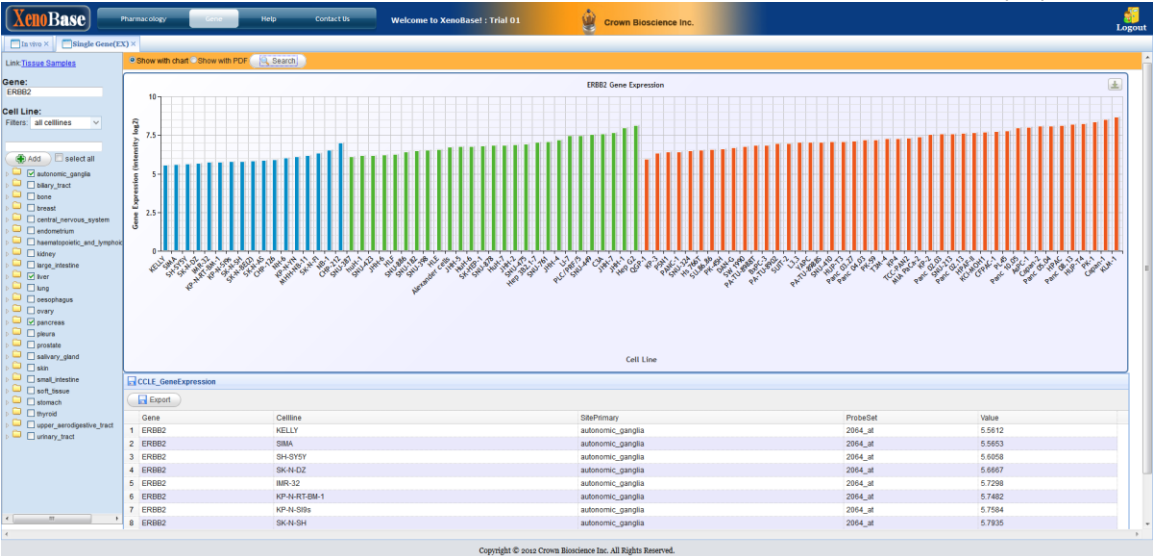
3. select all: you can also tick "select all" to select all the cell lines in the category.



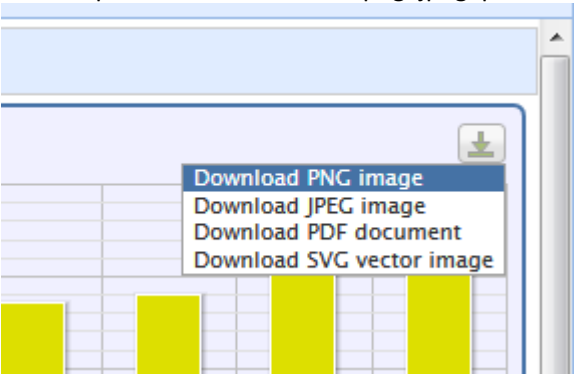
4.2 displaying result

After gene and cell lines are selected, click the button of , then the expression profiling of this gene is displayed:

1. When the number of selected cell lines is less than 200, the format below would be displayed:



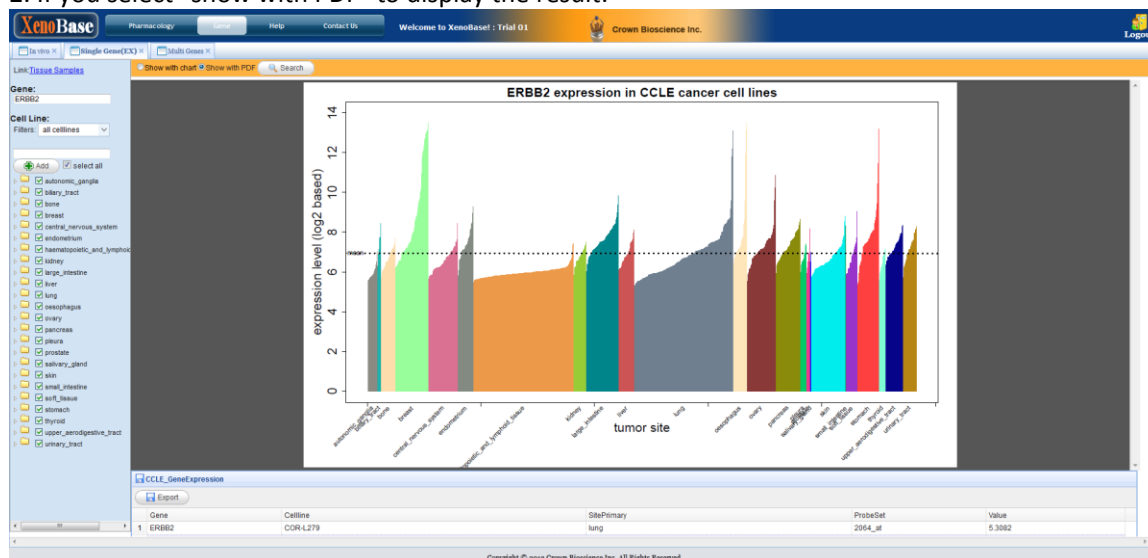
The chart can be zoomed in, or exported in the format of png, jpeg, pdf or svg.



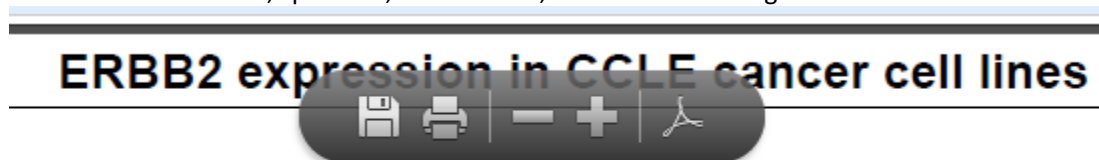
The table can be sorted in ascending or descending order of expression level by click "Value", It can also be sorted in alphabetic order by cellline:

Value ▾
12.7224
12.6289
11.9927
11.7533
10.9421
10.0053
8.5616

The table can be exported in xls format by clicking the button of  :
 2. If you select "show with PDF" to display the result:



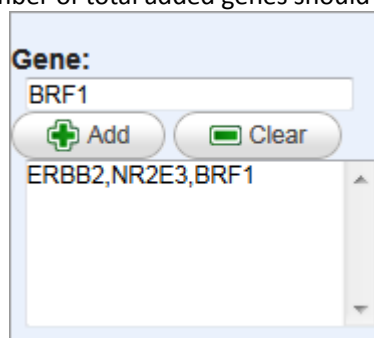
The chart can be "saved", "printed", "zoomed in", "zoomed out" using the PDF toolbar.



5. Gene-Gene Expression-Multi Genes

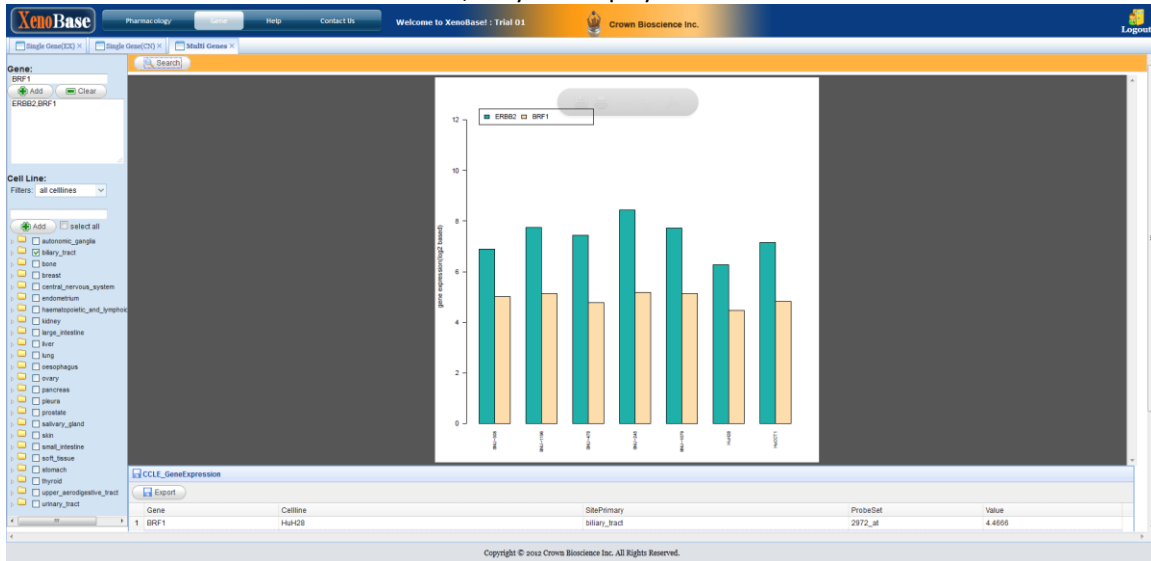
Click the "multi genes" item from the navigation bar.

Type in the gene name in the inputbox under the prompt of "Gene:" and click  to add genes to the select box (The number of total added genes should be no more than 8) .

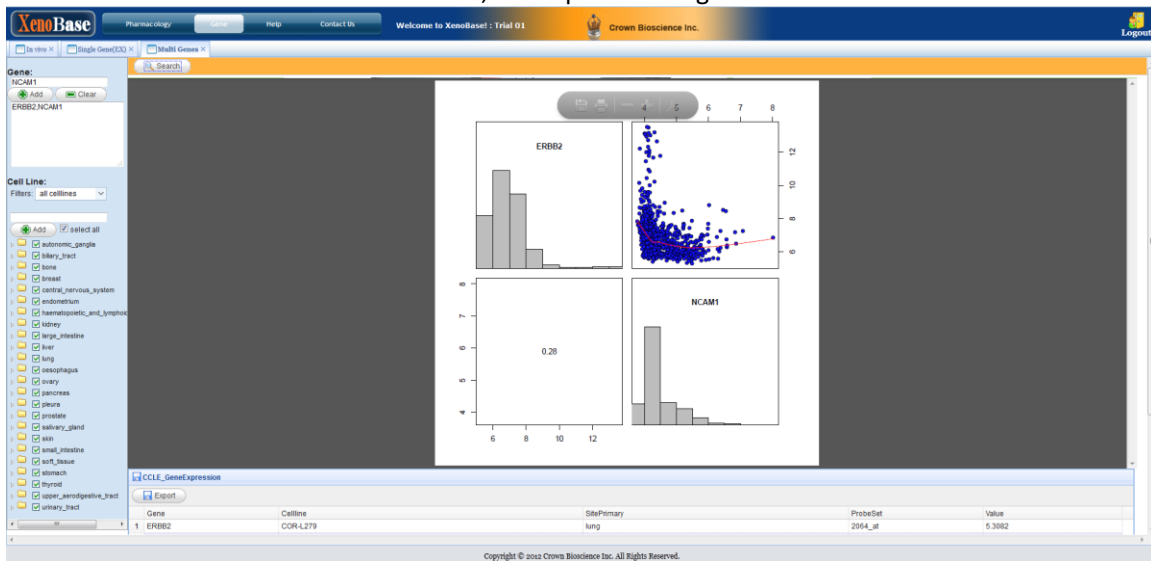


The screenshot shows the 'Gene:' input box in the XenoBase application. The input box contains the text 'BRF1'. Below the input box are two buttons: 'Add' and 'Clear'. Below the buttons is a list box containing the text 'ERBB2,NR2E3,BRF1'.

Select cell lines as previously described, click  to display chart and table. When less than 21 cell lines are selected, they are displayed in the form of bar chart:

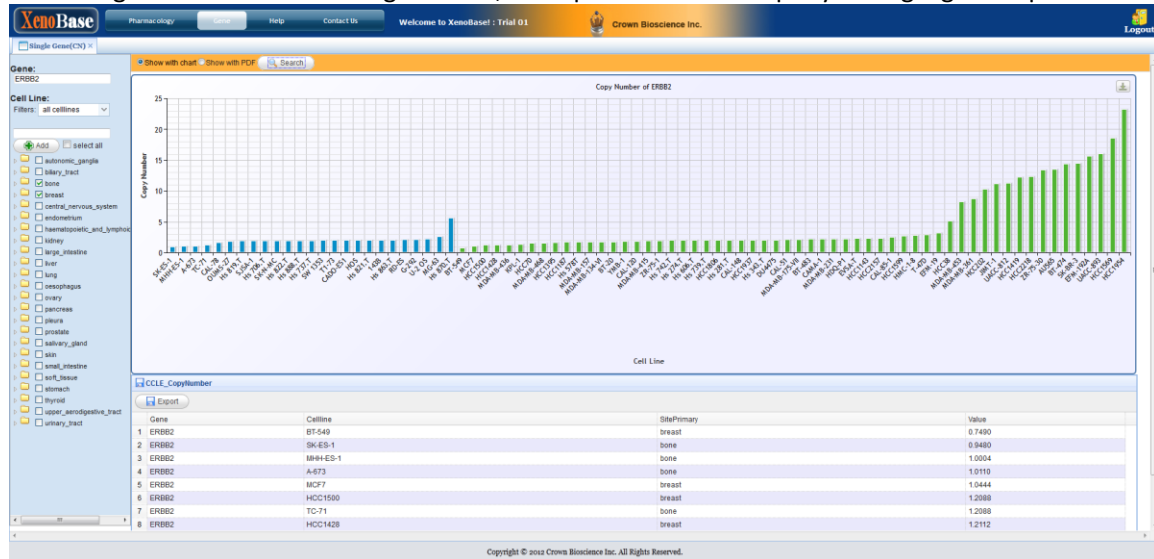


When more than 20 cell lines are selected, the expression of genes are in the form of matrix chart:



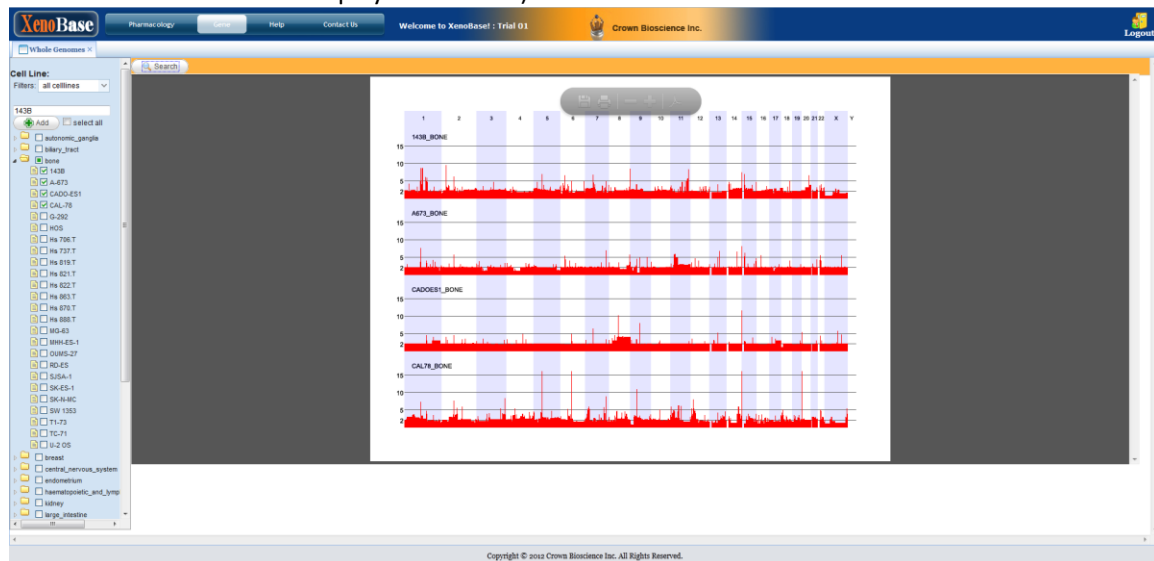
6. Gene-Copy Number-Single Gene

Click "Single Gene" from the navigation bar, and operate as for the query of single gene expression:



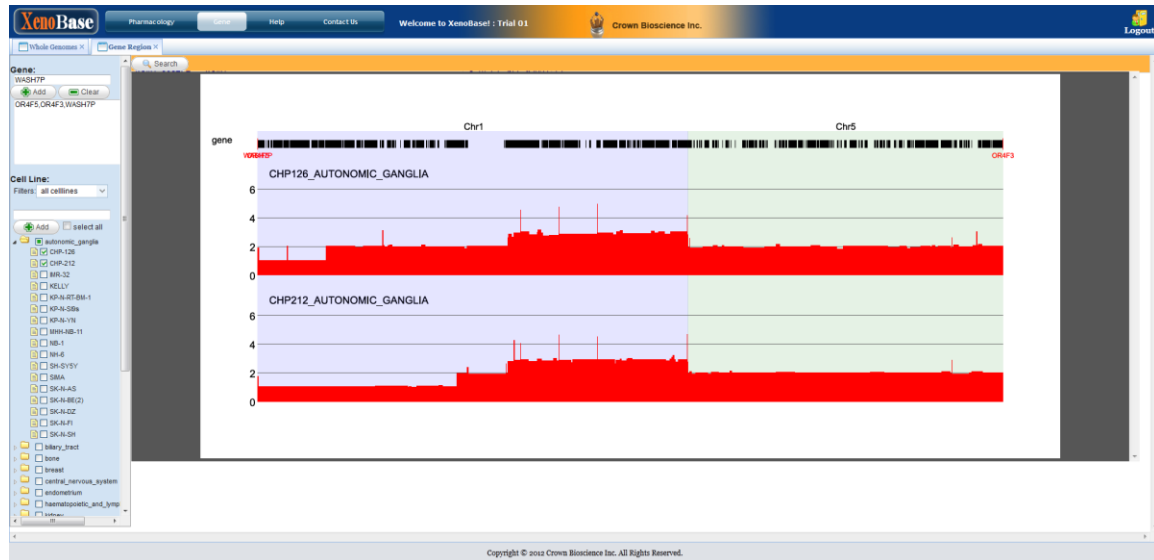
7. Gene-Copy Number-Whole Genomes

Click "Whole Genomes" from the navigation bar, select cell lines of interest, then click  to display copy number on the whole genome for the selected cell lines (At most 100 cell lines can be displayed at a time):



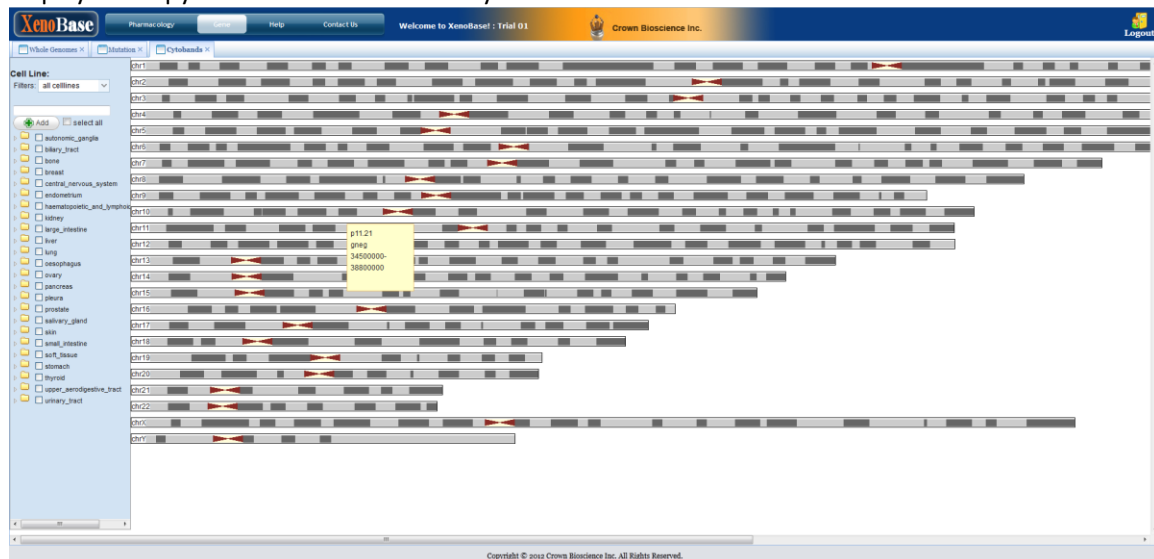
8. Gene-Copy Number-Gene Region

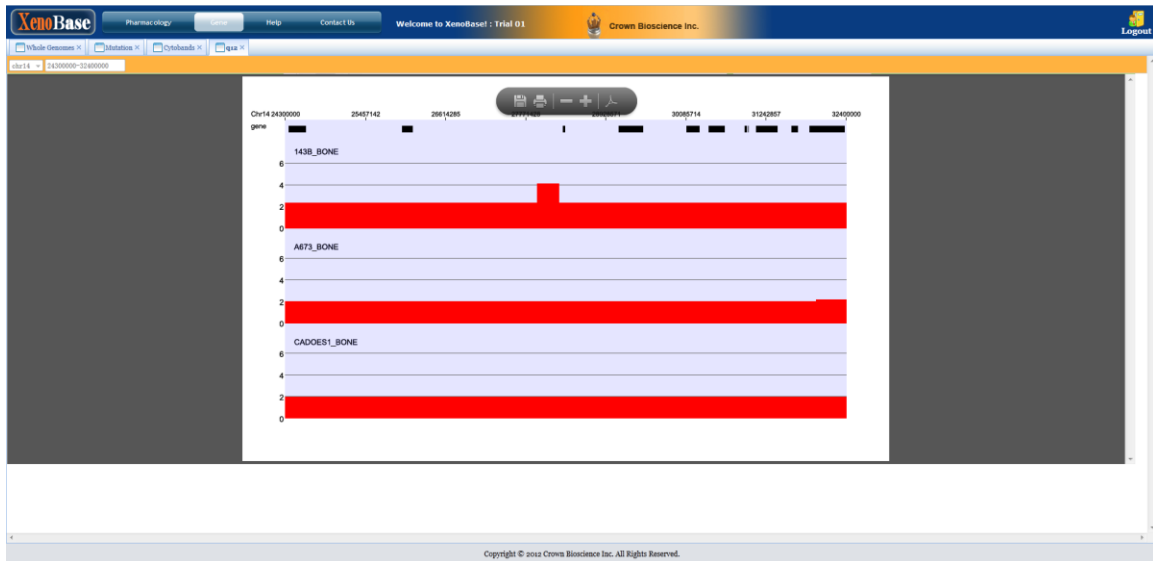
Click "Gene Region" from the navigation bar, select one or several genes and at most 250 cell lines, click  to display the copy number on chromosomes where selected genes are located at.



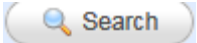
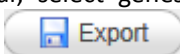
9. Gene-Copy Number-Cytobands

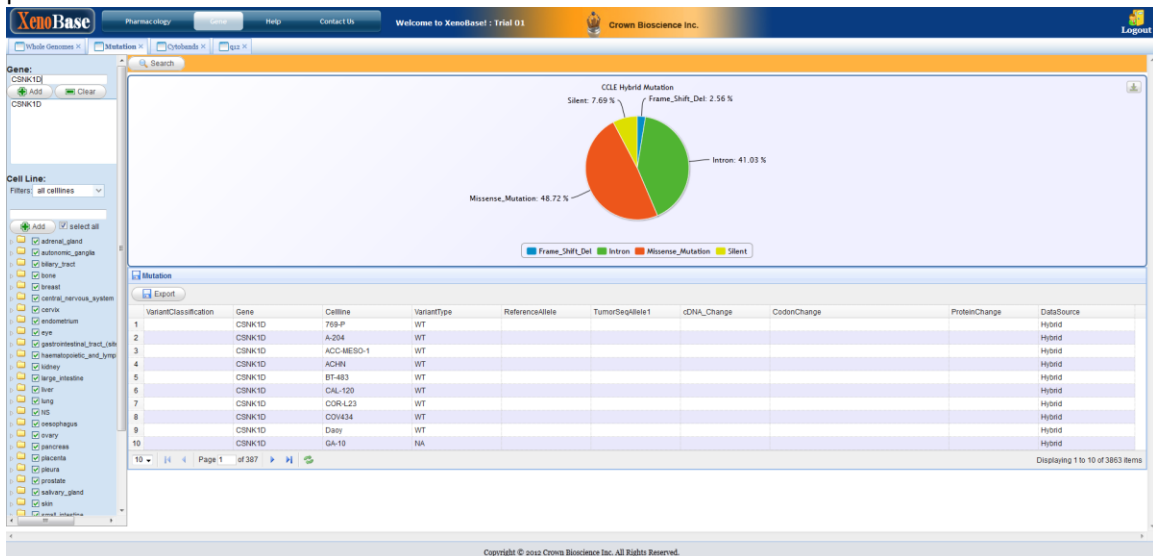
Click "Cytobands" from the navigation bar, and select at most 250 cell lines, click the cytoband bars to display the copy number of the selected cytoband.



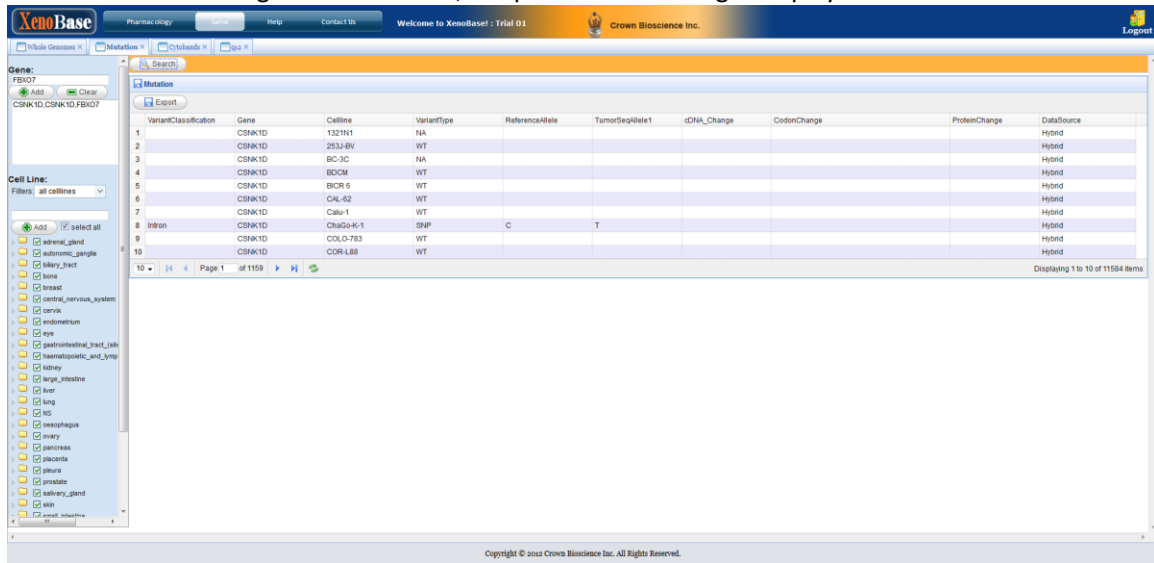


10. Gene-Mutation

Click "Mutation" from the navigation bar, select genes and cell lines, click  to display mutation status, and you can click  to export result in xls format. When only one gene is selected, the classification of mutations by CCLE hybrid method is displayed in pie chart:



When more than one gene are selected, the pie chart is no longer displayed:

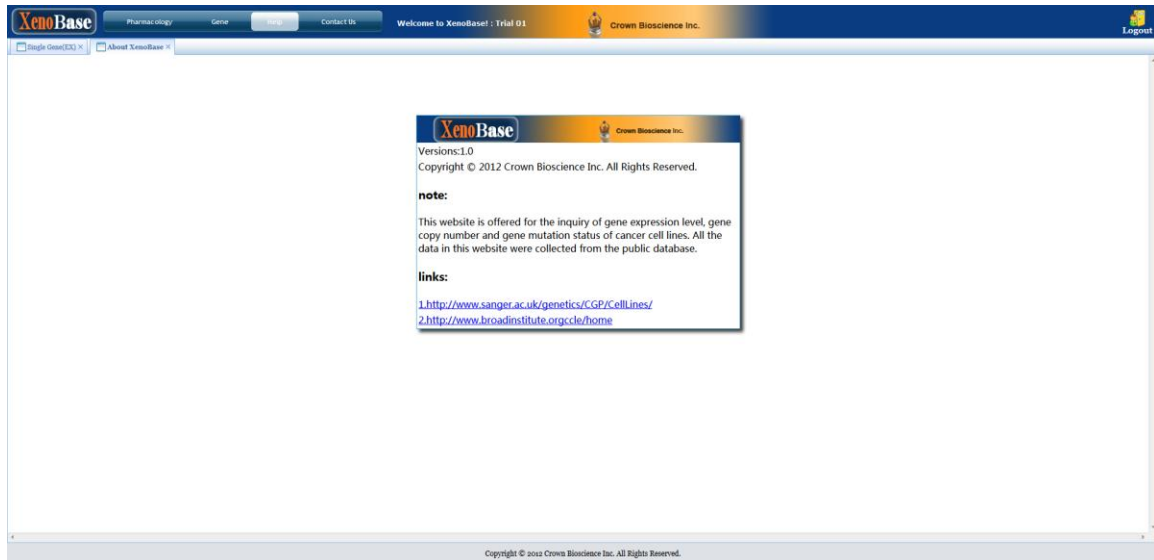


The screenshot shows the XenoBase website interface. The top navigation bar includes links for Pharmacology, Gene, Help, and Contact Us. The main content area displays a table of mutations for the gene CSNK1D. The table has columns for VariantClassification, Gene, CellLine, VariantType, ReferenceAllele, TumorSegAllele1, cDNA_Change, CodonChange, ProteinChange, and DataSource. The table lists 10 mutations, with the first 6 being WT (Wild Type) and the last 4 being SNPs (Single Nucleotide Polymorphisms). The sidebar on the left shows a list of cell lines and filters, including 'all cell lines' and 'select all'.

VariantClassification	Gene	CellLine	VariantType	ReferenceAllele	TumorSegAllele1	cDNA_Change	CodonChange	ProteinChange	DataSource
1	CSNK1D	1221M1	NA						Hybrid
2	CSNK1D	253J-BV	WT						Hybrid
3	CSNK1D	BC-3C	NA						Hybrid
4	CSNK1D	BIDM	WT						Hybrid
5	CSNK1D	BICR-6	WT						Hybrid
6	CSNK1D	CHL-62	WT						Hybrid
7	CSNK1D	Calu-1	WT						Hybrid
8	CSNK1D	ChaSe-K-1	SNP	C	T				Hybrid
9	CSNK1D	COLO-763	WT						Hybrid
10	CSNK1D	COR-L88	WT						Hybrid

11. Help-About XenoBase

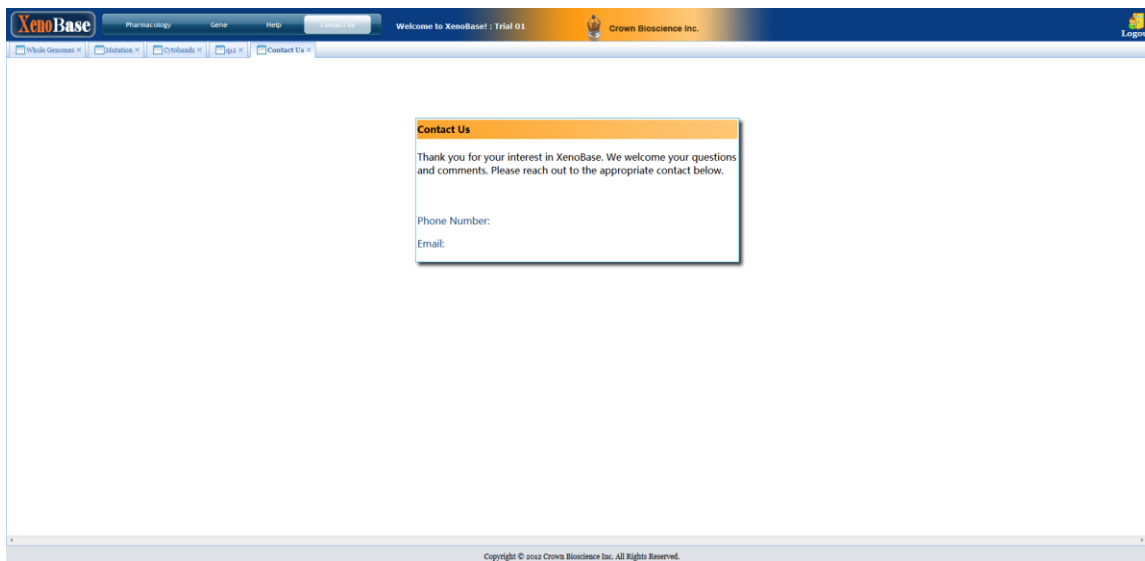
Click "About XenoBase" from the navigation bar, the page briefly describes the main content of this website and the resource used to create this website.



The screenshot shows the 'About XenoBase' page. The page contains the following information:

- Version:** 1.0
- Copyright:** © 2012 Crown Bioscience Inc. All Rights Reserved.
- note:** This website is offered for the inquiry of gene expression level, gene copy number and gene mutation status of cancer cell lines. All the data in this website were collected from the public database.
- links:**
 - 1. <http://www.sanger.ac.uk/genetics/CGP/CellLines/>
 - 2. <http://www.broadinstitute.org/cde/home>

12. Contact Us- Contact Us



13. Log out



Click the  at the top right corner to log out.